$g = 0$ to be approximately equal to the time duration from point of $g = 0$ to R.

10 **C** Given $\phi_Q = -2.0 \text{ J kg}^{-1}$

$$\frac{\phi_R}{\phi_Q} = \frac{PQ}{PR} = \frac{1}{4}$$

$$\phi_R = -\frac{1}{4}(2.0) = -0.5 \text{ J}$$

Work done = $m\,(\phi_R - \phi_Q)$ = (3.0) (–0.5 + 2.0) = 4.5 J

Work done is positive, since direction of external force is same as direction of displacement from Q to R.

11 **A** The defining equation of simple harmonic motion is $a = -\omega^2 x$ with a and x having opposite